RoboCupRescue Rapidly Manufactured Robot Challenge (RMRC) 2026 Rulebook

The details may still change between this document and the competition. There may also be things missing from this document. Please check https://rrl-rmrc.org for the latest version and submit comments to info@rrl-rmrc.org . Visit https://list.rrl-rmrc.org to join our mailing list and be kept up to date!

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Introduction

RoboCupRescue is a family of research competitions that drive the state-of-the-science in real world capabilities for robots used in applications such as search and rescue, hazardous materials response, and disaster recovery. It integrates with the NIST Standard Test Methods for Response Robots project, meaning that teams that do well in the competition have produced systems exhibiting capabilities that are immediately interesting to the real world end users who keep us safe every day.

This document describes the motivation, procedures, rules, and expected conduct for the RoboCupRescue Rapidly Manufactured Robot Challenge (RMRC). It is intended to be read in conjunction with the RMRC Arena Construction Guide. The latest versions of both this document, and the RMRC Arena Construction Guide, are available from https://rrl-rmrc.org .

Competition Spirit

The RoboCupRescue Rapidly Manufactured Robot Challenge (RMRC) combines open, real-world research challenges, with the rigorous, statistically significant, measurement science focus of the Major competition. The lower cost, more structured, and pedagogical nature of the RoboCupJunior Rescue competitions welcomes young robot researchers. It incorporates the structure and innovation, previously only available to well funded university teams, and makes it accessible to modestly funded high schools. The Rapidly Manufactured Robot Challenge combines incentives for learning about critical aspects of research, such as technical communication and collaboration.

Much like games such as golf, and as distinct from games such as chess, RMRC is a competition where all teams work together, against the application. Teams are rewarded both for performance against a wide variety of tests, as well as how they communicate their work to other researchers. Instead of individual holes of golf, teams perform different standard tests from the Standard Test Methods for Response Robots project. Unlike golf, however, teams are free to choose which tests they perform, the order in which they do them, and the number of times they try each test, to make sure that the result they achieve is reflective of their capabilities.

The scoring formula rewards excellence in commonly exhibited capabilities, as well as the demonstration of novel capabilities. Teams are further incentivised for developing novel systems that address real-world gaps in capabilities through additional Best-in-Class award categories.

Who is the Competition For?

RMRC is designed for teams of up to 6 students[1] and 2 mentors[2] (1 mentor for 3 or fewer students). Most teams consist of high school students, with teachers being the mentors. We also encourage teams to form outside of schools, such as in social groups and makerspaces. We particularly encourage university students to mentor groups of high school students, sharing their new-found knowledge and expertise.

NOTE: In some rare cases, such as when qualified teams pull out at the last minute, we may have enough space to admit additional team members and mentors. In general, we will only know this within a month or two of the competition. If this is the case, an announcement will be made on the mailing list at https://list.rrl-rmrc.org, inviting teams to apply to send more team members and mentors. Teams with more than 6 students and 2 mentors will be expected to make some team members available for volunteer duties at the competition.

Competition Summary

This section provides a very high level overview of how the competition is run. Detailed rules for the different parts of the competition appear in the rest of this document. All timetables are subject to change, please see the website and mailing list, available from https://rrl-rmrc.org for the latest details.

Qualification

For the International competition, qualification generally involves submitting a Team Participation Form (TPF) in response to a call for participation, and Qualification Team Description Materials (TDMs) a month or so after that.

The Qualification TDMs are judged by the RMRC Committee to determine team's readiness and qualification for competition. Some spots are generally held open for teams that qualify by winning domestic competitions and other regional qualification pathways.

Award winners of the World Championships in the previous 2 years (first, second, third, and Best-in-Class) are automatically qualified. They must still submit the Team Participation Form by the deadline to register their interest, and are encouraged to submit a Qualification Team Description Paper to tell the rest of the teams what they have done since their award.

[1] According to RoboCup Federation guidelines for RoboCup Junior students at time of writing, students must be 19 years of age or younger on the 1st of July of the competition year. Teams that do not satisfy this criteria may participate in the Major RoboCupRescue Robot League. In the International competition, students must also be at least 14 years of age on the 1st of July of the competition year.
[2] Mentors must be at least 18 years of age at the time of the competition (badge pickup) as they are expected to be the adult who is legally responsible for the team.

For Domestic competitions, these timings and requirements vary. Please monitor the mailing list at https://list.rrl-rmrc.org for details and variations for any given year. Contact info@rrl-rmrc.org if you wish to find out more information about domestic competitions that may exist in your area.

Updated Team Description Materials

Teams that qualify submit Updated Team Description Materials, due 2 weeks prior to the competition. These materials are scored by the RMRC Committee based on how well these materials help another team, new or otherwise, learn how to implement a particular RMRC-relevant capability. These materials must be released open source, under a permissive license. A growing library of these materials is available from https://rrl-rmrc.org . This score forms a multiplier onto the team's Preliminary Round score. This provides a strong incentive, particularly for the best teams, to also share their knowledge and experience, allowing other teams to build on their work.

In-person competition

The International competition itself generally consists of 1 setup day and 3.5 competition days. Timings may vary for domestic competitions.

Setup

During the setup day, teams arrive, register, set up their robots and equipment, help to set up the apparatuses, and practice within the apparatuses as they are completed. The first team meeting is held with mentors and team leaders; volunteers and Judges are identified, the timetable for the competition is announced, a safety walk-through is conducted, and any event-specific logistics and details are also shared with the teams.

Robots are generally photographed at this point for documentation and configuration management. As the competition separates out an overall real world mission into component tests, the robot hardware[3] should stay the same through the first 3 days of competition. For instance, a robot that has an arm to pick up an object should also be equipped with that arm when it traverses the terrain tests, as during a real mission it would have had to carry that arm over the terrain to get to where it needs to pick up objects. See the Robot Configuration definition for further details.

Preliminaries

Preliminary rounds are held during the first 2 competition days. Teams gain points by performing the various Standard Test Methods for Response Robots within the many different, separate, test method apparatuses that make up the arena. The precise rules and scoring methods for each test are described in more detail later in this document.

[3] In general, software updates are ignored in recognition that these robots are prototypical and on the premise that in a real scenario, it is conceivable that software can be changed over the air.

Many teams may compete in these different tests at the same time. Missions are 5 minutes long and start every 10 minutes, with each mission confined to one test apparatus and apparatus setting. Missions are run according to a global clock and all missions start at the same time, 6 times an hour.

This competition is team-run, meaning that all mentors are encouraged to volunteer and spend some time as a judge. With this help from our community, we can test multiple teams concurrently on different apparatuses. The tests are designed to be easy to score, with most simply a matter of counting laps. There is limited-to-no need for judgment calls, the robot's operator double checks and verifies the tally, and the scores are written on a publicly viewable scoreboard shortly after each mission.

Prior to each competition session, teams collaboratively determine their timetable for the following competition period (see "At the Competition: Preliminaries" later in this document). Given that there are so many opportunities to run, teams are not obliged to run in every time slot, and indeed are encouraged to allow time to troubleshoot, rest, and practice.

Only the best mission score that a team achieves in a given apparatus and setting counts towards the team's total preliminary score, so teams have ample opportunity to repeat tests if they choose to. The tests are designed so that within 5 minutes of robot run time, the best teams will have ample opportunity to perform a statistically significant number of iterations of the test. Teams do also have the opportunity to move time slots during the day, subject to availability, in response to unexpected issues or to take advantage of opportunities to improve performance.

The scores for each test are tabulated and normalized. For each test and setting, the best score(s) become 100%, with all other scores normalized accordingly. This balances the scores of easy and difficult tests, and encourages teams to develop new capabilities for tests that have not received as much attention.

Finals

The finals are held on the 3rd day but the preparation is done on the evening of the 2nd day, after the preliminary rounds have finished.

After the preliminary rounds, the scores are added up and combined with the updated Team Description Material multiplier. The largest gap in the scores between the teams in 3rd to 7th place is used as the cutoff for finals qualification, thus the number of teams admitted to the finals can range from 3 to 6 teams. This policy acknowledges the inherent uncertainty in the scoring process and aims to make the cut at a location where it is the most statistically significant.

Depending on the schedule, either the evening prior to the finals or the morning of the finals, the finalist teams collaborate in designing and building the finals arena by arranging the individual test method apparatuses into a continuous maze. Competition organizers serve to mediate between the teams to ensure that all finalist teams have input into the design.

The timing of the finals day is flexible and depends on the number of qualified teams. Depending on available time, and number of finalist teams, teams run 2-4 missions of around 15-30 minutes, with 5-10 minutes between runs for take down/set up of the next team. Teams will all have the same amount of run time available to them, robots will run one at a time, and have access to the full maze.Teams score one point for traversing an apparatus in one direction, and another point for traversing it in the opposite direction. Teams may traverse a given apparatus more than once in a given direction, but receive no additional points for doing so. Similarly, non-traversal tests, such as manipulation and sensing, can only be performed once per mission.

Team Challenge

The last day of the competition is only a half-day, due to the award ceremony. During this time, a casual “Team Challenge” competition is held, as a way for students, mentors, and organizers alike to have some fun and demonstrate collaborative robot capabilities in a less stressful manner.

Throughout the previous 3 days of the competition, the organizers solicit suggestions from teams for games or obstacles that may be played on this last day and a decision is made on the evening of the 2nd day, after the finalists are announced. This means that teams who did not qualify for the finals have a day to prepare and practice. The Team Challenge varies from year to year, based on the interests of the competitors. Unlike the other rounds, robots may be modified for this challenge.

Awards

Awards are given for first, second, and third place, based on the number of points achieved in the finals. Scores that are very close, generally within 5%, are considered ties, in acknowledgement of the inherent uncertainty of performance measurement. In rare cases, particularly if many teams are tied in the finals, organizers may use preliminary scores to break ties. Best-in-Class awards are also given, based on a subset of preliminary scores. The RMRC Committee may also issue an Open Source and Innovation award, and an award for the winner(s) of the Team Challenge.

How Does a Team Get Started?

RMRC has many features that are designed to make it easy for a moderately well resourced team to get started!

Building the Robots

Robots for RMRC are of a size that makes them easy to build using 3D printed and laser cut parts, and with components such as controller boards, cameras, and servos that are common and cheaply available on the hobbyist market.

There is a growing library of designs and tutorials, produced by teams over the past few years, that new teams can make use of to jump-start their development. See https://rrl-rmrc.org for details.

Building the Arena

The arena construction guide is available from https://rrl-rmrc.org and is designed to be compact and cheap to produce. Teams may start by building a single pallet and a few different terrain components that may be mixed and matched for testing, without the need to build a full arena. Teams in a local region are also encouraged to get together to form larger arenas and hold mini-competitions to practice.

Definitions

Apparatus

An apparatus is a piece of equipment that is used to perform one or more tests. An apparatus may itself have variations or settings that change depending on the test.

Arena

All of the competition space, including lanes and bays. Does not include administrative areas or the team paddock.

Bay

A single apparatus or set of apparatuses that supports one robot performing a test at a time. A bay may support one test (possibly with multiple settings), multiple tests that share an element (e.g., the dexterity tests which share a terrain), or multiple tests by swapping out terrains or other apparatus elements within one bay. This may be useful for smaller regional competitions and for teams building the apparatuses at home, where space and resources may be limited. Bays have a marked start position.

Competition Lead

The Competition Lead(s) will keep time and overall score (on a scoreboard visible to all competitors), without bias to any one individual or Team. They will ensure that fair play is observed, along with managing the paperwork aspects of the competition. They will also coordinate safety concerns (e.g. locating appropriate safety and fire equipment) and ensure teams, mentors, and judges are kept up-to-date on announcements.

Finals

Teams who access the finals (through their in-person scores during the preliminaries, TDM multipliers, etc), will have the opportunity to design and implement the design of the maze.

For the sake of fair play, it is strongly encouraged that **all** qualifying teams, who will be moving to the Finals, work together to decide and assemble their preferred configuration as a group, on the evening of the end of preliminaries. Once the maze has been assembled, this would allow for the greatest amount of time to practice and familiarize themselves with the maze.

Inspection

Generally held before competition begins, the Competition Lead(s) and/or Judges will determine if a robot configuration is safe, and follows the spirit and rules of the competition. An inspection

may be called for at other times, if the Competition Lead(s) and/or Judges determine there is a need for it.

Judge

Judges are volunteered from available Mentors, (badged) Parents, and adult competitors from the Major league. (If a Major team volunteers to assist in judging, ensure they are aware of their own competition schedule, and count the entire team as one judge, to ensure coverage in the event of Major chaos.) A judge will keep score, without bias, and deliver scores to the Competition Lead for input onto the scoreboard. Judges should avoid keeping score for their own team.

Lane

A bay that is long and thin, usually a Z-pallet (see construction guide), and is generally used for terrain and obstacle type tests.

Lap

Some tests involve the robot performing laps of the bay.

General definition:

1. Start in contact with one end wall during the mission time.
2. Traverse the bay without leaving the bay.
3. Make contact with the opposite end wall.
    a. If the robot does not make contact with the end wall before returning, both (incomplete) laps do not count.
    b. It is up to the operator to determine if they have made contact with the end wall before starting the next lap. They should not be informed of this by the judge, team-mates, audience, or anyone else.
    c. Contact with the end walls must be made with a part of the robot (including arms, wheels, and flippers). The tether or anything else connected to the robot only by a wire, string, or similar does not count.

First lap:

The first lap of the mission shall begin at the end wall marked as the start position. If the robot moves prior to the mission start time such that it is no longer in contact with the end wall, once the mission starts it shall be driven, or positioned by hand, so that it touches the starting end wall before continuing the mission.

Successive laps:

Successive laps shall be run in opposite directions.

Teleoperated Lap

By default, a lap is considered teleoperated. The judge will tally the lap in the teleoperated section of the score sheet.

A robot that is teleoperated may operate wirelessly or on a tether. The operator makes decisions based solely on the information that the robot delivers to the operator. The operator is the only person who controls the robot's movement and manipulation capabilities.

A lap that is performed with the assistance of autonomous behaviors, such as automatically centering in gaps, is still considered teleoperated if it does not fully satisfy the requirements of an “Autonomous lap” (below).

In the competition setting, the operator may not look at the robot or the bay, and may not communicate with anyone but the judge.

Autonomous Lap

For a lap to count as an autonomous lap, the operator shall not touch the controls or otherwise issue commands to the robot unless any part of the robot, including arms, wheels, and flippers, is within 30 cm of either end of the bay. Parts that fall off the robot and/or parts that are not solidly attached to the robot (e.g. the tether or anything connected to the robot only by a wire, string, or similar) do not count.

The operator may monitor the robot’s progress through the operator control unit only, without touching it. As usual, they may not look at the robot or bay, and may not communicate with anyone but the judge. It should be clear to the judge that they are not issuing commands to the robot (e.g., by having their hands well clear of the OCU).

The team shall announce that they are attempting a lap autonomously prior to beginning autonomous operation (or continuing autonomous operation in a new lap), and receive confirmation from the judge to ensure that they are aware and can confirm that the operator is hands-off the controls. If the team successfully completes the lap autonomously, the judge will tally the lap in the autonomy section of the score sheet **instead** of the teleoperated section.

A team may attempt a lap autonomously, and during the lap, decide to take over teleoperation outside these 30 cm areas. If they do so, the whole lap now counts as a teleoperated lap. The judge will tally the lap in the teleoperated section of the score sheet.

Resets

If a reset is called, the robot shall re-start from the starting position. See the full definition of Reset below for more information.

Maze/Labyrinth

Used in the Finals, the Maze comprises the available lanes/bays, and can be configured completely flat or tiered (with multiple levels, including the drop test), depending on the available space and the amount of challenge the teams wish to face.

Walls, including end walls, may be removed during the configuration of the maze, in order to integrate bays/lanes into one another.

Mentor

Each team may have up to two mentors to assist and supervise. These individuals will be badged and are strongly encouraged to assist with running the competition (be it as a volunteer judge, assembly of lanes, etc). They must be at least 18 years of age and be able to legally represent the team (e.g., sign consent forms).

Mission

This is the 5 minutes of active time within a time slot, when the robot is in operation.

Multiplier

Based on a variety of factors (TDMs, autonomous operation, etc) a team's score may be increased. This will affect a team's ability to move from preliminaries into the finals. Multipliers may apply per point (e.g., reading an individual Landolt-C), or to the entire preliminary score (e.g., the Updated TDM multiplier).

Operator Interface

The equipment, provided by the team, that the operator uses to control the robot. This includes the operator side wireless networking equipment (e.g., the access point or the network interface, but not including the network cable/tether).

Aisleways/thoroughfares will be kept clear of any tripping or fall hazards, including, but not limited to, the tethers, robot/robot parts, operational equipment, etc.

Operator Station

The location where the operator controls the robot, along with the necessary equipment and furniture (table/chair). The operator station is placed close to the lane to minimize the tripping hazard associated with cables reaching from the operator station to the robot, but far enough so that the operator can be positioned facing away from the lane.

Operators/pilots of the robots will not be able to have line of sight of the robot as the test is run. This is to more closely duplicate the real world setting and challenges faced by rescue robots. No-one apart from the judge shall communicate with the operator once their timed run has begun (e.g., to indicate the robot's location, whether a test was successfully completed, etc) apart from in an emergency, which will automatically end the mission.

Tethers may need to be run from the operator's station to the robot to allow for communications without needing to use wireless networking. One team member may be required to manage these lines, without disrupting the robot or pilot, and without causing a safety/tripping hazard to others in the arena. See “Rules for Each Test” for further details on tethers.

Preliminaries

Preliminary rounds/preliminary scores are trialed first at an in-person competition. Scores from the preliminaries will determine which teams will move onto the finals.

Penalties

All participants are expected to adhere to the intent of fair play. If this is not observed, penalties up to, and including, dismissal from the current competition, and barring from future competitions, may be meted out to individuals or entire teams. Further information is provided in the Code of Conduct.

Reset and Mini-mission

If anyone physically touches/moves the robot or arena during a run (eg. robot falls out of the terrain, robot gets stuck, tether gets pulled), a reset is called. This means that the robot is placed at the starting position of the bay, the scores for that run reset and then the run continues as if for a new mission, but only in the time that remains. Each ‘reset’ results in a separate ‘mini-run’ (or ‘mini-mission’ in RRL Major terminology). The judge shall record each “mini-mission” on a new score sheet. The score of the best ‘mini-run’ becomes the score for the whole run.

Robot Configuration

A robot configuration (often abbreviated to just “configuration” or “robot” elsewhere in this document) refers to the robot, along with any physical accessories and settings that require physical access to the robot to alter. A robot is expected to maintain the same robot configuration from the start of the preliminaries to the end of the finals (or end of preliminaries if the team does not reach the finals).

Motivation

The reason for this is because the competition consists of many individual tests that, together, represent a real mission. In a real mission, a person cannot touch the robot to change things. For instance, a robot that needs to pick up an object needs to have its arm installed as it traverses the terrain to get to where it needs to pick up the object. Therefore, in the competition, a robot that wishes to score points for manipulation must also have its arm installed for the mobility tests. Software changes are generally allowed, on the premise that in a real robot, this can often be achieved remotely.

Documentation

The configuration of the robot at the start of the competition is documented on, or prior to, the first day, through photographs, ideally with a 20 cm grid background for scale. These photographs can be referenced to check that the robot configuration has not changed. At minimum, the following angles should be taken of the robot in both its Stowed Posture (see below for definition) and a typical operating posture.

- Front, Side, and Top images.
- A “publicity” image that easily/readily identifies the robot (e.g., at an angle).
- A close-up image of unusual and special features of the robot, as declared by the team.

Changing configurations

A team may elect to change a robot configuration during the competition but this will be considered a separate robot configuration for the purpose of scoring. Thus unless a team does this early in the competition, they risk reducing the number of missions that they have to accumulate a good score to qualify for the finals. The robot configuration that the team runs in the finals must be the same as a robot configuration that accumulated a high enough score to qualify for the finals. A team may only enter one robot configuration in the finals.

Repairs

It is understood that sometimes it is unavoidable that a robot configuration may change during the competition, such as because the robot is damaged and it may not be possible to repair it to exactly the same capability. If the team wishes to have the repaired robot configuration considered to be the same as the original robot configuration, so that they can continue accumulating scores, they shall consult with the Competition Lead(s) to see if any additional modifications are necessary.

In general, the Competition Lead(s) will consider the repaired robot configuration to be the same as the original robot configuration if they are satisfied that the repairs have not provided an unfair advantage to the team going forward. The Competition Lead(s) may also decide that some past missions may count while others do not.

For example, if the team has performed manipulation missions, then breaks the robot’s arm and is unable to repair it:

- The team can keep their previous mission scores by satisfying the Competition Lead(s) that the robot would not score higher in subsequent tests in its broken configuration than its original configuration. This may be achieved by:
    - Keeping the broken arm on the robot, securing it as necessary in a shape that it was originally able to take. The team may adjust the shape of the arm between subsequent missions as long as the Competition Lead(s) are satisfied that those positions are ones that the original robot configuration could have achieved by itself without physical human intervention.
    - Replacing the broken arm with something of similar size, shape, and weight, and satisfying the Competition Lead(s) that it affects the robot in a similar way to the broken arm.
- If the Competition Lead(s) decide that the repaired robot configuration would score higher in subsequent tests than the original robot configuration, the Competition Lead(s) shall decide which previously performed tests can be counted, using the following guidelines.
    - Mission(s) that the Competition Lead(s) decide are unlikely to have been affected by the arm can be counted. For example, sensor tests (with sensors that are not on the arm, not blocked by the arm, or are otherwise still working).
    - Mission(s) that the Competition Lead(s) decide the repaired robot configuration is likely to outperform the original robot configuration can be counted. For example, mobility tests where the arm was not used to assist in mobility (e.g. by pushing against the ground or changing the center of gravity).
    - Mission(s) where the Competition Lead(s) decide that the repaired robot configuration is not likely to outperform the original robot configuration, or are unsure about, **can not** be counted. For example, manipulation tests.

NOTE: This rule also covers the complete replacement of the robot (which we consider to be equivalent to replacing so many parts that none of the old parts are part of the robot). If the new robot is similar in performance to the old robot prior to it becoming damaged, and otherwise satisfies the criteria above regarding "no unfair advantage" to the satisfaction of the Competition Lead(s), this is allowed.

Setting

Some tests may be modified, to increase difficulty. Generally, the initial form of a test will be the easiest.

Results are only compared for tests performed with the same setting. Tests with multiple settings are considered separately for the purpose of scoring. For the purpose of the competition, teams who attempt a harder setting should also attempt the test at an easier setting.

Each morning of the preliminaries, teams will have an opportunity to attempt the test at the easy setting before it is made more difficult. The timing of the change depends on scheduling and

team capabilities and is at the discretion of the organizers. This allows for additional challenges for the competitors, without taking up extraneous floor space during competition time.

Session

Each trial day is broken up into multiple sessions, typically two or three, with breaks for meals and lane inspections in the middle. These sessions are further broken up into 10 minute time slots.

Starting Position

Where appropriate, a specific location will be indicated in a bay or lane as the appropriate location for a robot to begin a mission. If in a lane, the robot will be required to begin its mission within physical contact of the wall at the end of the lane. If, by the start of mission time, it has rolled away from this starting position, it will be required to return to this position before commencing its mission. This may be done via teleoperation, or by hand (if it is running autonomously and has been programmed to avoid touching walls under its own power).

Stowed Posture

To score the autonomy multiplier for sensing and dexterity related tests, the robot must start its autonomous behavior from its stowed posture. The goal is to avoid teams teleoperating the robot and arms/flippers/etc. to almost the right position, then having the robot “autonomously” perform a trivial movement and gaining autonomy points.

We define a robot’s stowed posture as the posture where all articulated appendages that are directly used for sensing and dexterity (e.g., arms, legs, flippers, or variable chassis geometries) contribute the least to the robot’s 3D bounding box, as if it were stowed for transport. This does not include articulated components, such as wheels, that do not change the robot’s 3D bounding box as they move.

This posture will vary from robot to robot. In general, this will be the arm folded back or retracted against the body and sensor head tucked away, flippers folded back, and body in “sit” or “lie down” mode for legged robots. It is up to the autonomous behavior to un-stow the robot and perform the task. The Stowed Posture should be photographically documented at the start of the competition.

For some robots, the act of taking on the stowed posture may result in stability and safety issues. For example, the robot may be on an uneven surface and rely on flippers to stabilize. The team may elect to only use a subset of the robot’s articulated appendages to score a particular point, in which case this rule only applies to those appendages. For example, during the Keypad Omni test, a team may declare that they will only use the arm to push a key. They will only need to move the arm into the stowed posture prior to beginning the autonomous behavior but may keep the flippers where the operator left them. Any points scored by any other

articulated appendage pushing a key, such as a flipper, will not benefit from the autonomy multiplier until that appendage is also moved into its stowed posture.

Team Meeting

A team meeting will consist of at least one mentor per team, along with a student as team captain, along with the Competition Lead(s). Any safety concerns will be covered (fire, crush, explosion risks), announcements, etc will be covered. Volunteer Judges will also be recruited, and the team's runs scheduled for the following session.

Team Participation Form (TPF)

Supplied by teams as the first step in the qualification process. This supplies competition organizers basic team information and declares intent to compete.

Team Description Materials (TDMs)

Supplied by the teams, these papers are written and delivered prior to the in-person competition. Ideally, these will clearly demonstrate the team's process in preparing for the competition (e.g. equipment used, design choices made, challenges, etc).

These shall be written with the RMRC open source mind-set. The intent is two-fold: demonstrate the team's preparedness for in-person competition, and assist upcoming teams in future competition. (e.g. A poorly written paper, that can not be understood, about a "perfect robot" will help future rescue teams **less** than a comprehensible paper on a "poorly performing robot" that clearly demonstrates to future rescue teams what pitfalls they should avoid!)

The TDMs will be read and judged prior to the competition, on a variety of factors, and a multiplier score will be awarded based on these documents. Examples from previous years may be found at https://rrl-rmc.org .

Test

A test is a combination of an apparatus, procedure, setting, and metric that evaluates the performance of the robot in a specific capability.

NOTE: Prior to 2025, a Test was defined as "An apparatus, procedure, and metric that evaluates the performance of the robot in a specific capability." and that a "test" could have several settings. For example, a terrain test could have settings of "Horizontal" and "Crossover slope". These were considered the same test but scored separately. This caused some confusion as sometimes people didn't realize that they were scored separately. From 2025 onward, we will consider a test to also be defined by its setting. Thus, from 2025 onward, a terrain set to "Horizontal" is now referred to as a different test to the same terrain set to "Crossover slope".

Time Slot

Each time slot is 10 minutes; this includes 3 minutes for operator set up, 5 minutes of mission time, and 2 minutes of operator tear-down, after which the operator station must be clear so that the next team may begin setup. If a team is late, they may run in whatever time is left but they must end their mission 2 minutes before their time slot ends in order to ensure that the operator station is clear in time.

NOTE: Previously autonomy was granted extra time. This has been replaced by the 4x autonomy multiplier.

Work Area

This is the non-competition area, for competitor's and mentor's use. Tables, chairs, powerpoints, etc will be assigned to specific teams, based on space and equipment availability of the location.

Competition Logistics and Procedures

Before the competition

Qualification

All teams must go through the qualification process set by the organizing committee before the competition. It is typically conducted in the following order:

- Team Participation Form (TPF) submission
- Qualification Team Participation Form (TDM) submission
- Notification of qualification
- Updated TDM submission.
- RoboCup Competition and Symposium

Team Participation Form (TPF)

The first step in the qualification process is to complete and submit the TPF. Each team wishing to qualify must produce the following information. Note that this information (apart from contact information) may be released publicly.

- Team Name
- Organisation
- Country
- Contact person
- Email
- Telephone number
- Estimated number of students
- Estimated number of mentors
- Estimated number of robots

Please follow the specific and additional information required for each competition and year, as well as the method by which this information should be submitted to the organizing committee.

Team Description Materials (TDM)

All teams must share their resources during the qualification process; this competition is open source.

Participants must submit an updated TDM prior to the competition, which will be scored and included in the preliminary round score as a multiplier to the in-person competition scores to decide on the teams that will move on to the finals. The TDM describes both the team and the robot with the technical details. Note that this document will be publicly available.
The TDM should contain the following information:

- Logistical information
    - Team Name

    - Organisation
    - Country
    - Contact person
    - Team public contact email
    - Team website (if present)
- Introduction summarising
    - The team.
    - The technical aspects that it focuses on.
- Description of the system with the following components
    - Hardware
    - Software
    - Communications
    - Human-robot interface
- Description of the application
    - Setup and packing of your robot and operator station
    - Mission strategy
    - Experiments and tests that you have done or will do
    - How the particular strengths of your team are relevant to applications in the field.
- Conclusion, summarising
    - What your team has learned so far
    - What you plan on doing between now and the competition.
- Appendix
    - One table per robot outlining the components and estimated cost of your robot.
    - At least one picture, 3D rendering or technical drawing of your robot.
    - Be sure to highlight particular features of your robot.
    - A list of software packages, hardware and electronic components that you have used, or plan to use, particularly those from the Open Source community, through the Open Academic Robot Kit or otherwise.
    - A list of software packages, hardware and electronic components and designs that you have, or plan to, contribute to the Open Source community, through the Open Academic Robot Kit or otherwise.
- References (to other work you have used)

The TDM mirrors the Team Description Paper (TDP) goal from the Major competition, but in a form that allows for more flexibility, making it easier for students to participate. Teams may submit their materials in the form of a traditional TDP that follows the Major format (available at www.RoboCup.org). Specific information about deadlines and other TDM requirements can be found in the competition announcements at https://rrl-rmrc.org .

At the competition

Inspection

The robots will be inspected by a panel of judges before the start of the competition and at other times during the challenge to ensure that they meet the constraints described in these rules. Participants will be asked to explain the operation of their robot in order to verify that construction and programming of the robot is primarily their own work. Unlike in other Junior competitions, in RMRC, mentors are allowed to assist in the design and construction of the robots but the students still need to demonstrate both a working knowledge of the entire robot system as well as knowledge in at least one specific aspect of the robot's design, construction and/or operation.

All tests must be attempted with the robot in a configuration identical to what was initially inspected to be considered together. For instance, running with an arm for dexterity tests and then removing it for mobility tests will be treated as if the team entered a new robot. Teams who wish to make changes to their robot during the competition may request this of the Organizing Committee (OC) on the understanding that, especially if the change is not due to damage sustained during competition, this may be considered a completely different robot and reset all scores achieved up until that point. See the definition of "Robot Configuration" for more details.

Preliminaries

The preliminaries consist of each team running their robot(s) through the test methods multiple times and collecting points (per robot). Each trial is for a single robot; while teams may wish to bring multiple robots, they must be aware that this will split their available time slots. Entry into the finals is decided per robot, so this may reduce a Team's chances to enter the finals.

Time schedule

Each time slot of 10 minutes consists of 3 minutes preparation time, 5 minutes of run time, and 2 minutes to pack up.

The judge starts the run timer when the team leader declares that the team is ready, or at the latest after the 3 minutes of preparation time has elapsed. This happens regardless of whether the team is ready or present. Time slots are not delayed. The run will be stopped 8 minutes after the start of the time slot (or 5 minutes after the mission started if the team started early).

The testing schedule is determined at the team leader meeting prior to each session in the following way:

- Many teams run concurrently in separate test method apparatuses. This maximizes the amount of time available for teams to compete, demonstrate their capabilities and score points.

- During the team leader meeting, teams take turns nominating their run slots for the following day's preliminary competition and populating the scoreboard with their time slot nominations.
- In random order, teams take turns nominating a timeslot and test for a run.
- Nominations must conform to the following constraints:
    - Teams can only nominate once in each time slot.
    - The maximum number of nominations for a given time slot depends on the number of volunteer judges available.
    - Some apparatuses are shared between tests and thus a team nominating for one precludes another team nominating for the other in the same time slot.
- Nominations end when no team wishes to nominate for any more time slots or there are no more timeslots left that satisfy the above constraints.
- At any time during competition, up to 10 minutes prior to a timeslot, two team leaders may approach the OC and request to transfer or swap a timeslot. This will be noted on the scoreboard.
- At any time during the competition, up to 10 minutes prior to a timeslot, a team leader may approach the OC and request to swap the time slot to a different test, or add more runs in empty slots, as long as the apparatus is available and the above constraints are satisfied.
- Each time slot is 10 minutes and includes 3 minutes for operator setup, 5 minutes of operation, and 2 minutes to pack up.
- Teams will set up at the nearest Operator's station to their indicated test, with one team member acting as the operator of the robot. Once the test time begins, the only team members that will be in the area will be the robot operator and the team member needed to manage the tether.
- Due to the potentially congested radio environment and restrictions due to local regulations on wireless communications, it is strongly recommended that all teams have the ability to run with a communications tether.

Scores

- The goal is to score as many points as possible in the test methods. In general, you get one point for getting from the start zone to the end zone and another point for the way back from the end zone to the start zone (and so on). See individual test details for the scoring metric(s).
- At the end of each run, both the judge and operator must sign off on the score reached by the judge. Any concerns in scoring are to be raised immediately with the judge and a member of the OC.
- Other team members watching are encouraged to cross-check and verify the scores before the judge submits the score on the publicly viewable scoreboard. Scores will be tallied and recorded on the scoreboard, with the judge and robot operator signing the score submitted, to ensure accuracy throughout the competition. If there is a disagreement, do not sign the scoreboard, this denotes that the judges will need to revisit this score and make a decision.

- Every team can repeat a test, as often they want, subject to the aforementioned team, apparatus, and judge scheduling constraints. Only the best resulting score will be kept for the task and team.
- All scores will be normalized per test method, so that the best team in that test gets 100 points. This calculation is done after all teams have completed all tests in the preliminary round. The other teams get points proportionally. Example: For test method Dexterity 1: If team A scored 20 points and team B scored 10; then at the end of the preliminaries the score of team A will be set to 100 and the score of team B to 50. This way, for each test method the best team gets 100 points.
- This scoring method balances the difficulty level across differing tests. If a test is so difficult that only one team attempts it, the team will receive 100 points. Conversely, to score well in an easier test that everyone can do, teams must be close to the best.
- The sum of all points of all tests of a team is then multiplied by the updated TDP score to decide the qualification for the final round.

Finals

At the conclusion of the preliminaries, teams will be notified at a team leader meeting who has qualified for the finals and what factors went into this decision based on scoring as outlined in chapter “Preliminaries”. Note that qualification into the finals is per robot, not per team. Only one robot per team may enter the finals.

Following the announcement of the finals-qualified teams, members from each finals-qualified team will collaborate to design and build the arena from the lanes used in the preliminaries such that all competing finalists are satisfied with the layout. OC members will work with teams to ensure each team´s opinion is equally considered. Once the teams have finished assembling the final configuration of the maze **no further changes may be made** without the **agreed consensus** of all competing final teams.

Final scores are gained by traversing obstacles and terrains formed from the preliminary lanes, reconfigured into a single maze. This demonstrates the ability of a robot to get to where it needs to go to perform its task in a manner more representative of the typical challenges facing a rescue robot. Points can also be collected by sensor and dexterity tests. This represents the ability of a robot to perform its task. During the final, points can be collected both autonomously, where the autonomy multiplier is available, and in teleoperation. Apart from re-attempting for autonomy, all points can only be collected once per final mission. See “Rules for Each Test” for more details.

Until the end of the preliminaries, we won’t know how many teams qualify for the finals so we won’t be able to fix the number and duration of finals runs until then. Depending on the number of qualifying teams and the amount of time available, each team will have 1 to 4 runs, each run being around 15 to 30 minutes long. The 1st, 2nd and 3rd places are decided by the accumulated scores of finals runs, with preliminary scores only used to break ties (score differences of less than 5% are considered ties).
The procedure for each finals round is as follows:

- Before your mission, move your robot and other equipment to one of the provided staging tables. (If space permits, instead of a staging table, competition organizers may elect to have two Operator Stations, with one team running in one while another team sets up in the other.)
- 3 minutes Setup phase: The operator sets up in the Operator Station, facing away from the maze, as they were in the preliminaries.
- Run Phase: During the run phase, the robot starts at one end of any of the lanes in the arena and traverses through the arena from one lane to another. The robot scores one point for traversing each lane in each direction, for a maximum of two points per lane. (Subsequent traversals of the same lane, in the same direction, don't yield any additional points.) Teams should decide on their start point and proposed path through the maze carefully to best play to the strengths of their robot. The operator may draw and refer to a map of the arena during their run if they desire.
- If a reset occurs, the score is reset and the robot moves to one end of any lane nominated by the operator (it need not be the previous start point). As per the preliminaries, the best 'mini mission' is the one that counts.
- 2 minutes to clear the arena.

Note: If your robot has radio issues, a tether/cable to communicate with the robot is acceptable. In this instance you will be allowed a tether operator, however they are forbidden to communicate with the pilot during the run and must manipulate the tether without affecting the mobility of the robot. It is recommended that teams considering using tethers factor this into their input into the final arena design and into their planned route through the arena.

Team Challenge

A Team Challenge is offered as an optional challenge to finish off the international competition. Similar to the finals, the tests and tasks are developed and discussed together between the teams beforehand, and the participants should expect increased difficulty. In the Team Challenge, different teams are combined by lot, based on the number of teams interested in participating in the challenge, so that each team has an equal number of robots.

The Team Challenge is an optional, casual activity, as a way for students, mentors, and organizers alike to have some fun and demonstrate collaborative robot capabilities in a less structured manner.

Throughout the previous 3 days of the competition, the organizers will solicit suggestions from teams for games or obstacles that may be played on this last day and a decision is made on the evening of the 2nd day, after the finalists are announced. This means that teams who did not qualify for the finals have a day to prepare and practice. The Team Challenge varies from year to year, based on the interests of the competitors. Unlike the other rounds, robots may be modified for this challenge. It is recommended that the teams think about a strategy beforehand how they can help each other with their robots, as in this round, robots may push, carry or pull each other.

As the last competition of the event, this may be an interesting time to challenge team mates and robots. Feel free to get creative and have (safe) fun!

Trophies and Certificates

For the RMRC, trophies and certificates will be awarded to teams based on a number of categories. These are:

RMRC Overall

The following trophies result from the total score of the arena rounds:

- First Place

The following certificates result from the total score of the arena rounds:

- Second Place
- Third Place

Best in class certificates

The team/robot with the highest score in a specific robot class wins the according Best in Class certificate.

- Only the runs in the preliminary rounds count for Best in Class.
- To win any Best in Class certificate, you need a positive, non-zero score in 80% of the available tests.
- It is possible that a team could win more than one Best in Class certificate.

A certificate will be awarded to the winner in each of the following categories:

- Best in Class Mobility
- Best in Class Dexterity
- Best in Class Sensing
- Best in Class Autonomy

Note that the Open Source Innovation award in previous years has been replaced by the Updated TDM multiplier.

Team Challenge

All teams that have won the team challenge will be awarded with a certificate.

Rules for Each Test

NOTE: The rules should be read in conjunction with the Arena Construction Guide, available from https://rrl-rmrc.org , which provides more details as to the exact geometry of the test.

Introduction

There are a total of 18 individual tests available to teams in the preliminaries. Some of these tests are related and share the same apparatus and bay, but have different settings. Others share a bay because they share a common element, such as a terrain.

Most of the tests can be navigated with a mobile remote-controlled robot that has at least one sensor to transmit the environmental data to the operator (driver). The operators do not have eye contact with the robot. In general the operator is placed facing away from the robot.

The competition is divided into preliminary rounds and a final round. In the preliminary rounds, teams collect points by performing tests during missions. Teams may repeat any test, utilizing remaining time slots, to attempt a better score. Where a team performs the same test multiple times with the same robot, the best score is retained. The collected points of the preliminary rounds will be normalized, added up, multiplied by the Updated TDM score, and used to determine the final participants.

For each time slot, the team has 3 minutes to start up the robot and operator station in the operator booth of the respective test lane. After that, the team gets 5 minutes of mission time. At the end of the mission time, teams have 2 minutes to pack up and clear the arena.

For lap-based tests, the robot must be in physical contact with the start and the end wall to complete a lap. For each successfully completed lap, the judge scores one point. The conditions for successfully completing a lap are described below for each test, as are the conditions for successfully scoring a point on tests that are not lap based.

The robot must not be touched during its mission, nor moved by anything but its own power (e.g., a person pulling on the tether is not allowed but the tether is allowed to be secured to the arena at the start of the mission so that a robot equipped with a motorized tether drum or winch can choose to wind itself back). If the robot is touched, this counts as a reset, points are set to zero, and the test is re-started with only the remaining time. At the end of the 5 minute mission time, the largest contiguous score is taken.

List of Tests

This is the total list of the 18 tests currently defined in RMRC. The best-performing team in each test earns a normalized 100 points. Every effort will be made to provide all of these at the World Championships, however operational constraints may result in some being unavailable or modified. Regional competitions may run a subset of these tests.

- Incline and Center - Horizontal
- Incline and Center - Inclined
- Sand and Gravel
- Ramps on Crossover Slope - Continuous Ramps
- Ramps on Crossover Slope - Pinwheel Ramps
- Elevated Ramps
- Stairs
- Align
- Keypad Omni
- Linear Rail Pick
- Linear Rail Inspect
- Sensor Cabinet
- Labyrinth - Flat
- Labyrinth - K-Rails
- K-Rails - Horizontal
- K-Rails - Crossover Slope
- Hurdles - Single Layer
- Hurdles - Double Layer

Scores for tests with the same apparatus at different settings should not be combined, nor should they be weighted in an attempt to yield equivalent scores. This is because tests at different settings challenge different aspects of robot performance. For example, a given terrain apparatus flat might be a test of speed while the same apparatus set to crossover slope might require a slow, methodical approach.

Note, in the finals, teams may optionally choose to also run the Drop Test. This is not recommended for preliminary rounds.

Example Schedule and Scoreboard

Apparatus =>	Incline and Center			Ramps on Crossover Slope					Dexterity Field				Labyrinth		K-Rails		Hurdles	
Test =>	Horizontal	Inclined	Sand and Gravel	Continuous Ramps	Pinwheel Ramps	Elevated Ramps	Stairs	Align	Keypad Omni	Linear Rail Pick	Linear Rail Inspect	Sensor Cabinet	Flat	K-Rails	Horizontal	Crossover Slope	Single Layer	Double Layer
9:00 - 10:00	Team Leader Meeting and Arena Prep																	
10:00																		
10:10																		
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12:00 - 12:30	Lunch and Administrative Catchup/Redo																	
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14:30 - 15:00	Break and Administrative Catchup/Redo																	
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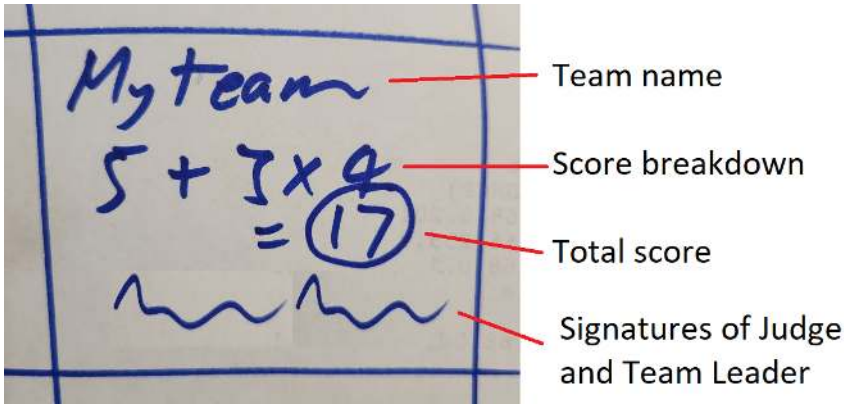
16:40																		
16:50																		
17:00 - 18:00	Cleanup, Team Leader Meeting, and Following Day Slot Assignment																	

Example timings for 36 timeslots per day. With 3 judges over 2 days, this provides 216 slots, which is enough for 10 teams to try all 18 tests with some buffer time. Note that some tests are mutually exclusive in a given timeslot as they share apparatus. Adjust according to available time, teams, judges, and apparatuses.



Example of a square in the table above, filled in on-site. The team name is written in during slot assignment (typically the previous evening) while the other fields (score breakdown, total score, signature and/or initials of judge and team leader) are written in after the run is complete. Where there are issues or discrepancies, notes can also be made (and signatures deferred until the issue or discrepancy is addressed). This allows anyone looking at the scoreboard to compute for themselves the current rankings of each team, including normalization (to the current best performer in each test).

In the interest of transparency, alterations should be crossed out. Should a box have insufficient space, a margin-note or foot-note should be made.

All of the scoreboards will be kept in a clearly visible location, to allow everyone to compare the scores and rankings. However, to reduce the foot traffic around the publicly displayed scoreboard, the Competition Lead(s) may post the scoreboards from previous sessions in a separate location from the current session, so judges may record scores and return to the lanes with minimal delay.

Once the judges have posted their tallies on the publically viewable scoreboard, the paper score sheets will be given to the Competition Lead(s) to keep in case there is a need for later verification.

Autonomy

For dexterity related tests, the autonomy bonus is intended for systems that sense the target objects and move accordingly. It is recommended that teams include evidence of the autonomous system's operation, such as overlays or console outputs, during operation so that the decision-making process is obvious. Systems based on training/replaying joint angles, teaching specific robot-relative points, or similar are not considered autonomy for the purpose of this multiplier (but teaching apparatus-relative points is OK because this requires good localization).
For a dexterity mission to be considered autonomously performed, the robot must not be teleoperated to within 30 cm of the object. The final alignment and approach to the object must be performed autonomously by the robot.

Incline and Center *(two tests)*

<u>Settings</u>

Note - these should be scored as separate tests, in separate timeslots. Teams are free to choose to do one without doing the other.

- Bay horizontal
- Bay inclined at 15 degrees

<u>Mission objective</u>

This lane will be used to test how well the robot can navigate through narrow spaces. This also shows how well the remote operator can perceive the environment via the robot's sensors.

<u>Lap preparations</u>

- Set the width of the bottleneck so that it corresponds to 120 % of the diagonal of the robot.
- The robot may start at either end and in any orientation.

<u>Scoring points</u>

- Completing a lap, defined as the robot starting in contact with one endwall and moving until it makes contact with the other endwall, without leaving the terrain, scores one point. Returning to the start point scores another point.
- The robot continues scoring points in this manner until the mission time ends.

<u>Autonomy bonus</u>

- A lap, from contact to contact, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate between teleoperated and autonomous laps in the same mission. For example, in an inclined bay they may elect to teleoperate up the incline for 1 point, and run autonomously down the incline, for 4 points.

Sand and Gravel on Crossover Slope

Mission objective

This lane will be used to test how off-road capable the robot is in outdoor environments.

Lap preparations

- Smooth the sand and gravel surfaces. Make sure that both ramps have adequate coverage of at least 25 mm.
- The robot may start at either end and in any orientation.

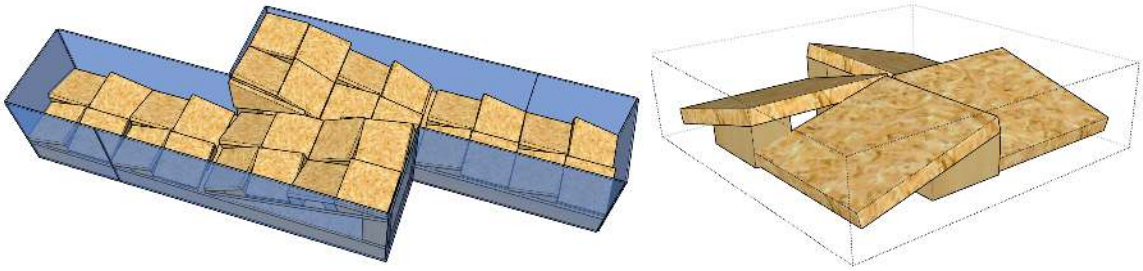
Scoring points

- Completing a lap, defined as the robot starting in contact with one endwall and moving until it makes contact with the other endwall, without leaving the terrain, scores one point. Returning to the start point scores another point.
- The robot continues scoring points in this manner until the mission time ends.

Autonomy bonus

- A lap, from contact to contact, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.

Ramps on Crossover Slope (two tests)



<u>Settings</u>

Note - these should be scored as separate tests, in separate timeslots. Teams are free to choose to do one without doing the other.

- Continuous ramps (Bay horizontal)
- Pinwheel ramps (Bay horizontal)

<u>Mission objective</u>

This lane also tests the off-road capability over uneven terrain. The transition between two oppositely inclined levels is particularly challenging.

<u>Lap preparations</u>

- The robot may start at either end and in any orientation.

<u>Scoring points</u>

- Completing a lap, defined as the robot starting in contact with one endwall and moving until it makes contact with the other endwall, without leaving the terrain, scores one point. Returning to the start point scores another point.
- The robot continues scoring points in this manner until the mission time ends.

<u>Autonomy bonus</u>

- A lap, from contact to contact, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.

Elevated Ramps

<u>Mission objective</u>
This lane simulates stepped terrain.

<u>Lap preparations</u>
- The robot may start at either end and in any orientation.

<u>Scoring points</u>
- Completing a lap, defined as the robot starting in contact with one endwall and moving until it makes contact with the other endwall, without leaving the terrain, scores one point. Returning to the start point scores another point.
- The robot continues scoring points in this manner until the mission time ends.

<u>Autonomy bonus</u>
- A lap, from contact to contact, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.

Stairs

<u>Mission objective</u>
This lane simulates a staircase. A simple configuration could be set up with low step heights, e.g., half the height of a standard step.

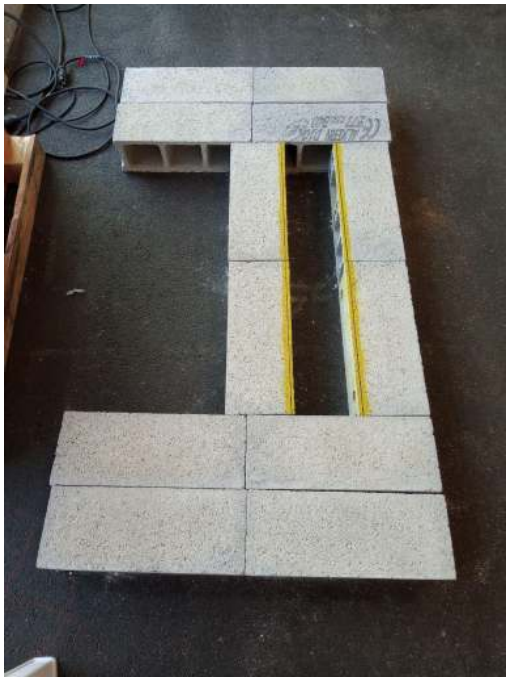
<u>Lap preparations</u>
- The robot starts on the ground at either end and in any orientation.

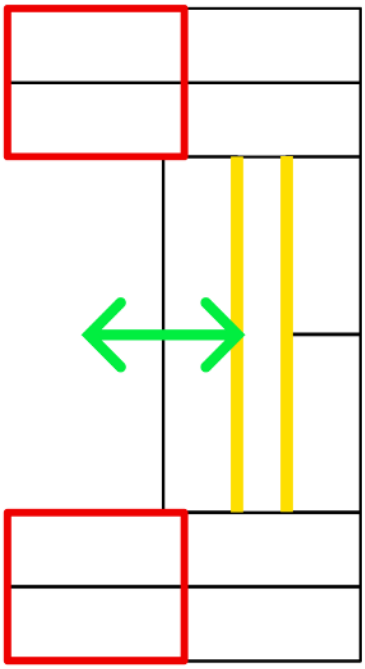
<u>Scoring points</u>
- Moving from any level, to any other level, is worth 1 point.
- A robot is considered to have moved from one level to another if it moves from only being in contact with one level, to only being in contact with the next level, without returning to the previous level for at least 5 seconds.
    - This excludes situations where the robot “bounces” up and down as it attempts to grip the edge of the step as it climbs, momentarily losing contact with the previous level, and situations where the robot momentarily “hangs” on one level without losing contact with the previous level.
- For example:
    - Driving up the stairs, touching each level on the way up, from the ground to the top level, is worth 3 points in total.
    - Driving off the top level directly to the ground (without touching the other levels on the way down) is worth 1 point.
- The robot continues scoring points in this manner until the mission time ends.

<u>Autonomy bonus</u>
- A step up, from all ground contacts on one step to all ground contacts on the next step, completed without any operator intervention, scores 4 points, as long as the robot stops by itself before any operator intervention. An operator intervention to stop the robot also removes the autonomy bonus from all steps since the operator gave the last command.
- A team may start a step autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.

- Teams may alternate teleoperated and autonomous steps in the same mission.

Align





<u>Mission objective</u>
This lane presents a challenge to the navigability of the robot. It must be driven over a simulated abyss with centimeter precision.

<u>Lap preparations</u>

- Adjust the gap of the abyss as shown by the green arrows so that the wheels / tracks / legs of the robot rest with only one centimeter of ground contact with the test apparatus on each side. The colored lines may be used as a guide.
- The robot starts in the red highlighted area at either end and in any orientation.

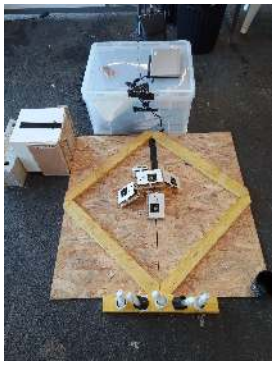
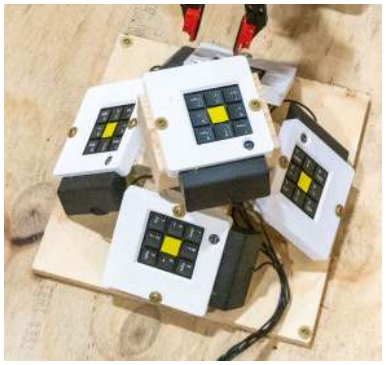
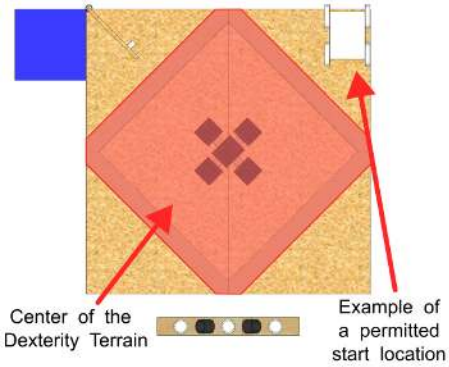
<u>Scoring points</u>

- Drive from one highlighted area, across the abyss, to the other highlighted area. For the purpose of this test, a robot is inside a highlighted area if all ground contacts are inside that area.
- The robot continues scoring points in this manner until the mission time ends.

<u>Autonomy bonus</u>

- A lap, from area to area, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.

Keypad Omni

<u>Mission objective</u>
This apparatus tests the robots' ability to precisely touch and push on objects in different orientations and locations, without touching things around the desired object.

<u>Preparations</u>
- Ensure that the scoring computer recognizes the keystrokes and that NUM LOCK is on (e.g., by pushing a key and making sure the corresponding number appears).
- Place the robot on one of the outer corners of the dexterity field, in a location and orientation of the team's choosing as long as it does not touch the beams of the terrain.

<u>Scoring points</u>
Each point may be scored multiple times per mission.
- The robot may drive anywhere on the dexterity field. Incidental contact with any part of the keypad omni (e.g., with a wheel) is fine.
- A key is considered pushed if the key's number is registered by the scoring computer.
- The robot may use any part of itself to push keys (including flippers, wheels, etc.).
- 3 points are scored for pushing the "5" key on a keypad (marked with yellow tape/paint) without pushing any other key on that keypad.
- Only 1 point is scored if any of the keys "1", "2", "3", "4", "6", "7", "8", and/or "9" are pushed (individually or in combination with the "5" key). See example below for clarification.
- A robot may return to score using the same keypad multiple times, but they must drive outside the center part of the dexterity field (so that they no longer touch the beams) before returning to the same keypad. In effect, driving outside the center part of the dexterity field "locks in" their score up to that point and allows them to score all of the keypads again. Failing to do so may *lower* their score (see example below).
- Example 1:
    - The robot pushes the "5" key on the top keypad, slips, and also pushes the "2" key, scoring 1 point.
    - The robot drives a short distance and then immediately pushes only the "5" key on the top keypad. They score no additional points as they have stayed in the center of the dexterity terrain since they last pushed a key on the top keypad.

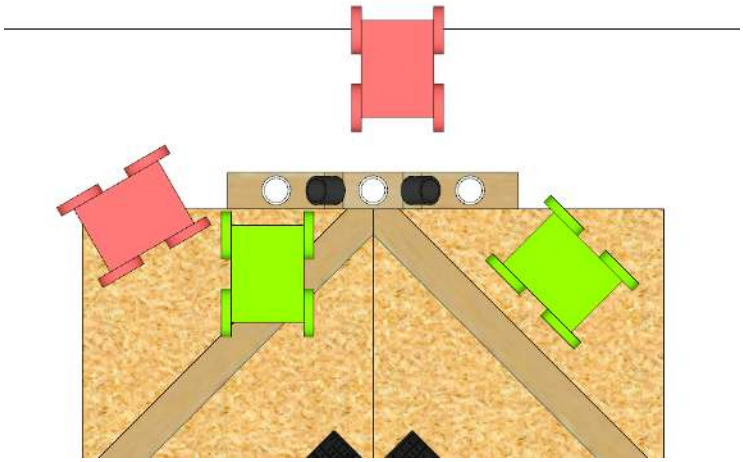
    - They drive out to a corner of the dexterity field, to the point that they no longer touch the beams.
    - They return to the top keypad and push the “5” key (without pushing any other key on that keypad), scoring 3 points before the mission ends.
    - Their total for the mission (before multipliers) will be 1 + 3 = 4 points.
- Example 2:
    - The robot pushes the “5” key on the top keypad, scoring 3 points.
    - The robot drives a short distance and then (possibly while maneuvering to push a key on a different keypad) pushes the “2” key on the top keypad. What would have been 3 points will now be reduced to 1 point.
    - If they score no other points before the time runs out, their total for the mission (before multipliers) will be 1 point.

NOTE: Only the number keys “1” to “9” count in this test. All other keys on the keypad are ignored, regardless of if they are covered or not. It is recommended that the NUM LOCK key be protected to avoid accidentally turning off NUM LOCK.

<u>Autonomy bonus</u>

- If the robot starts in its stowed posture and scores the point(s) with no operator input apart from a single command (e.g., a button push or mouse click), a 4x multiplier is applied to those point(s). The operator may drive the robot anywhere prior to issuing this command.
- The operator may take over to drive the robot, arm, etc., between point(s), but the robot must be in its stowed posture before the next attempt at an autonomy bonus.
- The operator may take over during an attempt at pushing a key to score points without the autonomy bonus.
  NOTE: In this test it is possible for an operator takeover to reduce points. For example, if the robot autonomously pushes just the “5” key on the top keypad (scoring 3 x 4 = 12 points), and then the operator takes over and accidentally pushes the “2” key on the same keypad before driving out and returning, the score for the top keypad will be reduced to 1 point (as they lose the multiplier and the 3 point score for cleanly pushing the “5” key). It is therefore recommended that teams’ autonomous behaviors end with all parts of the robot well clear of the keypad that was just pushed.
- Teams may alternate between teleoperated and autonomous points in the same mission.

Linear Rail Pick



Mission objective

This apparatus tests robots' ability to manipulate objects in different orientations and locations.

Preparations

- Place a manipulation object in each tube. See the Construction Guide for details of the manipulation objects.
- Place the designated container in any location on the field (not touching the linear rail).
    - The designated container shall be provided by the team and be no more than 30 x 30 cm in footprint and 10 cm high. (A cardboard box is fine.)
- Place the robot anywhere inside (but not touching) the beams of the dexterity field.

Scoring points

Each point may only be scored once per mission.

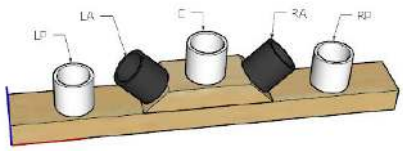
- The robot must be completely on the dexterity field (the OSB sheet and beams) to score any points. In the diagram above, the green robots are OK, the red robots are not.
    - The robot should not make contact with the ground outside the dexterity field (e.g., no driving to the other side of the linear rail).
    - The robot can overhang the edge of the dexterity field as long as no part of it touches the ground.
- Only the robot may move the manipulation objects.

- The manipulation object must be grasped to be manipulated. Techniques such as skewering, suction, static, adhesive, or magnets are not allowed.
- A manipulation object that ends up in the designated container scores 2 points.
- A manipulation object that ends up anywhere else scores 1 point.

<u>Autonomy bonus</u>

- If the robot starts in its stowed posture and scores the point(s) with no operator input apart from a single command (e.g., a button push or mouse click), a 4x multiplier is applied. The operator may drive the robot anywhere prior to issuing this command.
- The operator may take over to drive the robot, arm, etc., between point(s), but the robot must be in its stowed posture before the next attempt at an autonomy bonus.
- Between picking up a manipulation object and placing it in the designated container, the operator may pause autonomy and drive the robot as long as they do not move the arm. They may only issue a single command to resume autonomy.
- The operator may take over at any point to score the remaining points of that tube without the autonomy bonus.
- Teams may alternate between teleoperated and autonomous points in the same mission but the manipulation object must have been picked up autonomously for the drop to also be scored autonomously.

Linear Rail Inspect



<u>Mission objective</u>
This apparatus tests the robots' ability to inspect fine detail in different orientations and locations.

<u>Preparations</u>
- Ensure that each linear rail tube is empty.
- Place the robot anywhere inside (but not touching) the beams of the dexterity field.

<u>Scoring points</u>
Each point may only be counted once per mission.
- As for Pick, the robot must be completely on the dexterity field to score any points (i.e. no contact with the ground outside the dexterity field).
- Teams may attempt a tube multiple times, both teleoperatively and autonomously, but only the highest score in that tube (including multiplier) counts.
- Orientations are reported relative to the 2 character identifier (**not** relative to the top of the robot or top of the image). In clockwise order, these are: Up, Up-Right, Right, Down-Right, Down, Down-Left, Left, Up-Left.
    - All of the outermost Landolt-Cs in the linear rails will be in the "Up" orientation as the identifier is marked in its gap. This represents an "easy" point for getting the camera in a location where this is visible.
- In the preliminaries, each Landolt-C correctly read is worth 1 point, for a maximum of 25 points (5 tubes, 5 Landolt-Cs per tube).
- In the finals, only the 3 and 4th Landolt-Cs are counted, for a maximum of 10 points.
    - The largest 2 Landolt-Cs (including the aforementioned "easy" point) are not scored.
- To speed up the scoring, operators need only report the orientation of the smallest Landolt-C that they can read. Points will be awarded for the larger Landolt-Cs in the same tube. If they are wrong, they may re-attempt the next larger Landolt-C.
- To avoid questions of memorization, the operator must also show the judge the clear gap in the Landolt-C that they claim to be able to read.
    - If there is a disagreement, a screenshot (internal or with a cellphone/camera) shall be taken and the image evaluated by the judges on-site for a final ruling.

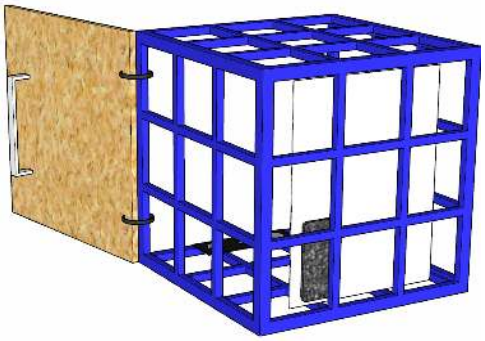
<u>Autonomy bonus</u>
- For a 4x multiplier, the operator shall:

    - Drive the robot to any desired location on the dexterity field.
    - Place the robot in its stowed posture.
    - Issue a single command at which point the robot (arm and base if necessary) must move autonomously to place the camera at the inspection location, stop, and indicate that the Landolt-Cs are ready to be read, and present a single, stationary image to the operator.
    - The operator then reads the Landolt-Cs from the stationary image on the screen without issuing any commands (such as adjusting the camera or zooming in).

  NOTE: This is scored differently from the autonomy multiplier in the sensor test as the challenge in this test is to move the camera to a location where it can clearly see the Landolt-C targets, rather than actually doing the reading. Therefore, in this test it is not necessary to autonomously detect and report or mark the directions of the Landolt-Cs.
- The operator may take over to drive the robot, arm, etc., between tubes as long as they place the robot in its stowed posture before attempting autonomy again.
- The operator may take over or return to the same tube to score remaining Landolt-Cs in the tube teleoperatively but each Landolt-C only counts once.
    - For example, if a team reads the largest 2 Landolt-Cs in tube C1 autonomously (4 + 4 + 0 + 0 + 0 = 8 points), and also reads 5 Landolt-Cs in tube C1 teleoperatively, either before, after, or by taking over from autonomy (1 + 1 + 1 + 1 + 1 = 5 points), their total for tube C1 will be 11 points (4 + 4 + 1 + 1 + 1).
- The operator may score a tube teleoperatively (or initially autonomously and then take over teleoperatively) and then, later in the same mission, re-attempt the same tube autonomously but they must drive the robot at least 30 cm away from where it was when they previously attempted that tube (and then driven back) before re-attempting it autonomously.
    - This prevents an operator from scoring a tube teleoperatively, saving the joint angles, stowing the arm, and then simply replaying the joint angles.

Sensor Cabinet

Formerly known as the Sensor Crate



<u>Mission objective</u>

This apparatus tests the robots' ability to gain access to and inspect a variety of different objects. Two different versions of a box are shown. The box may vary depending on the venue.

<u>Preparations</u>

- Ensure that the various sensor tests are present and correct: Hazmat label, Landolt-Cs, Thermal source, Magnet. An example of an appropriate sheet is available here.
- Door fully open or fully closed as desired by the team.
- Place the robot anywhere inside (but not touching) the beams of the dexterity field.

<u>Scoring points</u>

Each point may only be scored once per mission.

- 1 point for opening the door (wide enough to score at least one other test inside the cabinet).
- 1 point for correctly identifying the exact hazmat label out of the set provided. The location needn't be provided.
    - This may become more detailed in future years.
- 1 point for correctly identifying the orientation of the 3rd ring (approx. 1.8 mm gap) of the concentric Landolt-C.
- 1 point for correctly showing the location of the thermal source.
- 1 point for correctly identifying the North/South pattern of 3 disc magnets, 25 mm diameter in size, arranged in a line, with the centers 50 mm apart.

<u>Autonomy bonus</u>

For a 4x multiplier, the following must be true:

- The operator may drive the robot normally from the start point or the previous scored point but must place the robot in its stowed posture prior to issuing the command to start the autonomous behavior.
- The autonomous behavior must be started by the operator giving a single command (e.g., a button push, mouse click, command line, etc.).
- The operator must not give any more commands to the robot until the points for the test have been scored.

    - The robot may score multiple different points autonomously without operator intervention.
    - If the operator intervenes, points already autonomously scored retain their 4x multiplier, subsequent points are scored without the 4x multiplier (until the operator re-attempts autonomy).
    - If the operator wishes to re-attempt autonomy, they must first return the robot to the stowed posture.
- To autonomously score a Landolt-C (or multiple Landolt-Cs at the same time), the robot must stop autonomously when it has read the Landolt-C(s), provide a signal to indicate that it has read the Landolt-C(s), and report the readings by either:
    - Stating the direction(s) of the Landolt-C(s), relative to the identifier (which indicates up), e.g., by a screen overlay next to the Landolt-C or on a status line, or:
    - Mark the gap(s) on a single still image that is taken by the autonomous behavior. There must be no false positives. (Teams should not present a video with markers flickering in and out, nor should they present multiple images.)

  NOTE: This is different from the way autonomous Landolt-Cs are scored in the linear rail as the challenge is sensing itself, rather than moving the camera to the correct location.
- To autonomously score the thermal point, the robot must stop autonomously when it has found the location of the thermal source and report the location and the letter (if feasible) by a symbolic overlay on the camera image (e.g., a bounding box, a crosshair, or similar). A raw thermal image without annotation doesn't count.
- To autonomously score a hazmat label, the robot must stop autonomously, indicate to the operator that it has found a hazmat label, provide a single still image with an indication of where the hazmat label is (e.g., by a bounding box, crosshair, or similar), and the identity of the hazmat label (e.g., by an overlay or a status line). There must be no false or multiple detections.
- The operator must declare to the judge which sensor test(s) they are attempting before issuing the command to start the autonomous behavior.
- Teams may alternate between teleoperated and autonomous points in the same mission.

Labyrinth *(two tests)*

<u>Lane settings</u>

Note - these should be scored as separate tests, in separate timeslots. Teams are free to choose to do one without doing the other.

- Flat terrain
- K-rail terrain

<u>Mission objective</u>

This lane simulates an exploration task. Besides navigating through a meandering maze, some recognition tasks can also be solved. The recognition of QR codes, the inspection of a Sensor Cabinet and the possibility to register a map of the lane give additional points.

<u>Preparations</u>

- Check that all QR code sheets are still undamaged and attached to the walls. All QR codes should be unique. Sheets are available from here.
- Check whether all tests in the Sensor Cabinet are functional/recognizable.
- Fully open or close the door of the Sensor Cabinet as desired by the team.
- Place the robot on the ground at the entrance to the labyrinth on the opposite end from the Sensor Cabinet (up to but not touching the wooden floor panels).

<u>Scoring points</u>

Each point may only be scored once per mission.

- Correct automatic reading of a QR code gives one point.
    - Multiple readings are ignored (only one point per QR code).
    - The information from the QR codes must be written to a file and given to the judges at the end of the run.
    - During the run, the operator shall also declare each time a QR code is read and point it out on the screen to the judge. Multiple QR codes may be read at one time.

    - Each QR code on a sheet is considered as a separate point but each code must be read separately. (Reading the 1 mm code does not result in automatically being awarded the points for the rest of the codes on the page, for instance.)
- Correctly automatically placing the QR code in a map of the arena, such that its straight-line distance from the two closest fiducials (in real life) is accurate to within 30 cm, gives four (additional) points for that QR code.
    - The fiducials must be recognizable as (at least) ½ of a circle when seen in the map from above.
- Scoring points in the Sensor Cabinet - see “Sensor Cabinet”.

Note: To emphasize sensing, mapping, and autonomy, points are not awarded for just completing a lap of the labyrinth. Instead, teams should drive wherever they need to within the apparatus.

<u>Autonomy bonus</u>

- For this year, the mapping bonus replaces the autonomy bonus for QR codes (as the task of driving through the labyrinth to collect the QR codes isn’t actually that much of a challenge). We may re-evaluate this in future.
- The autonomy bonus still applies to the sensor cabinet as normal.
- Points scored autonomously in the sensor cabinet and QR codes accurately mapped contribute to Best-in-Class Autonomy, in acknowledgement of the importance of mapping to autonomous behavior.

K-Rails *(two tests)*

Settings

Note - these should be scored as separate tests, in separate timeslots. Teams are free to choose to do one without doing the other.

- Horizontal (Flat setting)
- Crossover slope

Mission objective

This lane tests the mobility characteristics of the robot by requiring slanted beams to be crossed along with overhead angled obstacles.

Lap preparations

- The robot may start at either end and in any orientation.

Scoring points

- Completing a lap, defined as the robot starting in contact with one endwall and moving until it makes contact with the other endwall, without leaving the terrain, scores one point. Returning to the start point scores another point.
- The robot continues scoring points in this manner until the mission time ends.

Autonomy bonus

- A lap, from contact to contact, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.

Hurdle Terrain *(two tests)*

<u>Settings</u>

Note - these should be scored as separate tests, in separate timeslots. Teams are free to choose to do one without doing the other.

- Single layer tube configuration
- Double layer tube configuration (Double step height / Two tubes mounted one above the other)

<u>Mission objective</u>

This lane tests the mobility characteristics of the robot by requiring it to negotiate slippery tubes to reach a different height level.

<u>Lap preparations</u>

- The robot may start at either end and in any orientation.

<u>Scoring points</u>

- Completing a lap, defined as the robot starting in contact with one endwall and moving until it makes contact with the other endwall, without leaving the terrain, scores one point. Returning to the start point scores another point.
- The robot continues scoring points in this manner until the mission time ends.

<u>Autonomy bonus</u>

- A lap, from contact to contact, completed without any operator intervention, scores 4 points.
- A team may start a lap autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.

Drop Test

(strongly recommended to be withheld until Final rounds)

Mission objective

Demonstrate the robustness of the robot. Survive, and continue to be capable of operation, after a sharp drop.

Lap preparations

- For the finals, an opening can be made in a raised lane (on any side of the 30 cm section of either end of a lane) for the robot to exit through.
- The drop can be placed as an endpoint within the maze, or can be placed to be used as a short-cut.

Scoring points

- Exit through the opening, land, and continue operation afterwards, scoring at least one point in another test before the end of the mission.
- 1 point for surviving a 15 cm drop.
- 2 points for surviving a 30 cm drop.

Autonomy bonus

- There is no autonomy bonus for the drop test as this is purely a durability challenge.
- There is no control challenge (the robot just needs to drive forward off the ledge).
- In normal operation, an autonomous behavior is supposed to avoid situations such as the drop test.

Final Maze

On the 3rd day, the best performing teams in the Preliminaries participate in the Finals. See “Competition Logistics and Procedures” for more details.

<u>Mission objective</u>
Demonstrate the robot’s various capabilities in a closed environment, which more closely resembles the challenges of a real world rescue operation.

<u>Preparations</u>

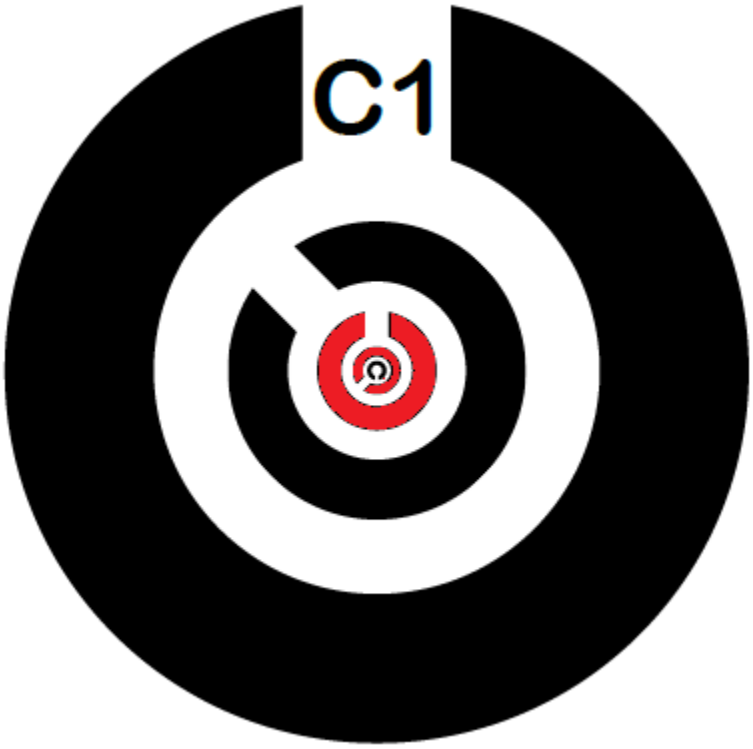
- The teams who have qualified will work collaboratively to construct the maze using all of the apparatuses.
- Mentors may assist (e.g. with power tools, ensuring fair play and safety, etc) but the design decisions should be the work of the competing team members.
- The final maze can be configured as a single layer (as seen above) or be configured into a two level maze.
- If the teams decide to attempt the Drop Test as a part of the final maze, the safety of all judges and team members must be considered as far as planning for a landing zone. (e.g. Avoid hard/sharp corners in the landing zone, as lithium battery punctures pose a fire risk!)

- In general, all apparatuses should be set to the hardest setting that any team was able to complete in the preliminaries.
- Exceptions and modifications (e.g., the balloons and wooden beams in the stairs shown in the example above) may be proposed to make the competition more interesting. These must be agreed to by all teams in the finals.

<u>Scoring points</u>

The finals requires teams to run the tests in sequence, without normalization. The following modifications are made to the scoring of each test.

- For traversal based tests (Stairs, align, and those in Z-Pallets):
    - Robots start anywhere in the maze and score one point for traversing each terrain or obstacle in each direction for a maximum of 2 points per terrain or obstacle per mission.
    - No points are awarded for running the same terrain or obstacle in the same direction subsequent times, apart from the Autonomy bonus - see below.
- For Dexterity Field tests:
    - All points can only be scored once per mission.
    - For all Concentric Landolt-C targets, only the 3rd and 4th largest Landolt-C optotypes count, and are worth 1 point each. These are shown in red in the example below.



    - For the Keypad Omni:
        - Only pushing the “5” key on each keypad counts and is worth 1 point.
        - Each keypad can only be scored once per mission.

        - Pushing any other key on the same keypad results in 0 points for that keypad (even if the “5” key was already pushed).
    - For the Linear Rail:
        - The two diagonal pipes (and optional the three vertical pipes) will be populated with manipulation objects, these must be removed before scoring the Landolt-C targets inside these two pipes.
        - One point is awarded for removing a manipulation object, and a second point is awarded for placing it into the designated container as per the rules for the Preliminaries.
    - For the Sensor Cabinet(s):
        - The door, hazmat, thermal, and magnet points are scored as per the Preliminaries.
- For the Labyrinth:
    - Only 5 code pages will be placed through the arena.
    - Only the 2mm code is scored.
    - Each code can only be scored once per mission (for a maximum of 5 points for the whole labyrinth per mission, before mapping bonus).
    - The mapping bonus replaces the autonomy bonus, as per the Preliminaries.
    - The sensor cabinet is not used (because there is already a sensor cabinet on the dexterity field).

<u>Autonomy bonus</u>

- The autonomous score of 4 points will be given in the same manner as the Preliminaries (including modifications, e.g., becoming a mapping multiplier for the Labyrinth).
- A team may start an individual terrain or obstacle autonomously and, if it fails, take over manual control and complete it without the autonomy bonus.
- Teams may alternate teleoperated and autonomous laps in the same mission.
- Teams may score a point using teleoperation and re-attempt the same point autonomously. Successfully completing the point autonomously results in the 4 points replacing (not adding to) the 1 point that was scored teleoperatively. The same restrictions as in the Preliminaries apply (e.g., the requirement to move away and come back in to re-attempt the Keypad Omni).

Team Challenge

On the final day, just prior to the award ceremony, all teams have the option of participating in the Team Challenge. See the “Competition Logistics and Procedures” section for more details. Multiple teams work together in a superteam to solve unique and fun challenges. Depending on the number of teams who wish to participate and the number of people in each team, it is suggested that each superteam be made up of between 2 and 4 teams.

While the Team Challenge is offered as an optional, casual challenge, the difficulty of the obstacles will increase from the challenge of the primaries.

Similar to the finals, the tasks are developed and discussed together between the teams beforehand.

The Team Challenge varies from year to year, based on the interests of the competitors. Unlike the other rounds, robots may be modified for this challenge, and full contact between the robots is allowed, including pushing, carrying or pulling each other.

The assignment of teams to N superteams is performed according to the following algorithm.

1. Teams are ordered according to their score in the preliminaries, after all multipliers.
2. Each of the top N teams are randomly assigned to a superteam.
3. The process is repeated with the next N teams until no teams are left.
4. The Competition Lead(s) may step in to re-balance teams if necessary to ensure that each superteam has roughly the same number of people (e.g., if the number of teams is not evenly divisible by the number of superteams, or if a superteam happened to end up with most of the large or small teams).

Code of Conduct

Spirit of Competition

It is expected that all participants (Competitors and Mentors alike) will respect the aims and ideals of RoboCup as set out in our mission statement. The Competition Lead(s), Judges, and other officials will act within the spirit of the event to ensure the challenge is competitive, fair, educational and fun.

Remember that the aim of this competition is for **all** Competitors to be competing against the **challenge.** In a real disaster situation, if multiple teams show up to help, they should **work together** to do the **best job possible.** This is an **open source** competition where we aim to all better ourselves and each other; please share and help each other grow!

Expectations

- All participants should review the most up-to-date rules of the competition (via http://list.oarkit.org/ and mailing list) before the competition, to ensure that everyone is participating fairly.
- All participants **must** wear **their own** badge, or be escorted by admin staff, while in the competition area. Badges **may not** be lent **or** traded between participants.
- All participants **must** wear close-toed shoes and secure any loose items/entanglement hazards.
- All participants should present at the venue early on setup day for Team Leader meetings, registration, arena walk through, rules clarifications, schedule updates, announcements, cover location-specific safety regulations, radio restrictions, etc.
- All participants **must** move safely through the entirety of the RMRC area (no running, climbing on equipment, obstructing footpaths, or other activities that could lead to the harm of self or others.)
- Competing teams have priority over tables, bays/lanes, and associated thoroughfares.
- **Only** Competition Lead(s), Judges, and Admin staff shall communicate with the Robot Pilots in preparation for, and during a competition run.

Competition Lead

- The Competition Lead(s) will keep time and overall score, without bias to any one individual or Team.
- At least one up-to-date copy of the Rules will be printed and available for clarification.
- At least one copy of the most recently submitted TDMs will be printed and posted for general review.
- Provide enough score sheets for the Judge's use.
- If there is any dispute, the Competition Lead(s) will make the final decision, with the appropriate committees brought in (as time and resources allow).

- At the beginning of each Session, the Competition Lead(s) will call for volunteer Judges and Team Leaders to gather and schedule their runs.
- At the end of each run, the Competition Lead(s) will receive and record scores from Judges.
- Ensure that **all** Judges, Mentors, and Team Leaders are familiar with safety risks and locations of firefighting equipment (e.g. buckets of sand, fire blankets and extinguishers for class A, B, C and D).

Judges

- At the beginning of each Session, the Competition Lead(s) will call for volunteers to serve as Judges.
- The number of Volunteer Judges will dictate how many Teams may run concurrently.
- (Recommended to reduce the number by 1 or 2, e.g. 9 Volunteers means 7 or 8 Judges, so that there is always a spare Judge to assist.)
- Judges may be volunteered from Mentors, (badged) Parents, and adult competitors from the Major league. (If a Major team volunteers to assist in judging, ensure they are aware of their own competition schedule, and count the entire team as One Judge, to ensure coverage in the event of Major chaos.)
- A Judge should avoid scoring for their own Team.
- Judges will keep score, without bias.
- At the end of each Run, the Judge will verify the score with the Team's Pilot (who will sign/initial scoresheet).
- Once the score is verified, the Judge will bring up the score sheet to the Competition Lead(s).
- (Recommended for the Judge to initial these scores on the scoreboard, along with the scoresheet, both to track scores if there is a question, and for the Judge to verify that the correct number was recorded.)

Mentors

- Each Team may have up to two badged Mentors assisting in the competition.
- **Only** registered, badged Mentors are allowed in the competition area.
- Mentors are encouraged to assist in operation of the challenge (e.g. as judges, suite manufacturer, scoring, etc).
- Mentors are permitted to build, repair, or be involved in the programming of their Team's robots in a limited capacity. (However, the Robots must be **mainly** the Competitor's own work!)
- Mentors may **not** interfere with Robots, Teams, or Judges.
- Supervise students; there shall be **no** unsupervised students in the work or competition area.
- Ensure that **all** equipment directly connected to power outlets follows event-location safety standards (rated appropriately, tagged as necessary, etc.)

- Ensure that **no** items are left charging/connected to power outlets **unsupervised** or **overnight.**

Competitors

- Each Team will be composed of up-to 6 students.
- Competitors should be capable of explaining their role within the Team.
- Competitors **must** be familiar with inherent safety risks of the Team's Robot (e.g. pinch, crush, burn, fire, explosion risks, etc).
- Ensure that there is **at least** one Mentor from your Team if any member of the Team is in the work or competition area.
- At the end of each run, verify the score with the Judge and sign the score sheet.
- **Any** dispute **must** be brought to the Competition Lead(s) before the next Team Leader Meeting/Session (at which point, all scores on the scoreboard will be considered finalized.)
- At the end of each run, Team will **remove** Robot and associated equipment in a safe, but efficient, manner, to allow the next Team to set up.
- Do **not** distract or interfere with other Teams or Judges while they are competing.
- Do **not** enter other Team's work areas, unless explicitly invited to do so.

Robots

- Any Robot that appears to be identical to another Robot, or appears to be factory standard, may be prompted for re-inspection.
- Tethers **must** be managed in a safe manner (e.g. avoid tripping hazards, damage to the arena, etc). Event-organizer-approved tape, carpet squares, etc may be utilized.
- Robots **will not** deliberately or repeatedly damage or destroy any testing apparatus.

Parents

- **Unless** a Parent is a Badged, Registered Mentor, they **shall** remain outside the competition and student work area.
- Parents shall **not** interfere with, or distract, **any** Team.

Guests

- Guests and Audience members are **welcome** in the viewing area.
- Guests and Audience members are **not** allowed in the Team work area or competition area.
- Guests and Audience members shall **not** interfere with, or distract, **any** Team.

Penalties

Rules will be enforced at the discretion of Competition Lead(s), Judges, officials, competition organizers and local law enforcement authorities. These individual(s) will decide if the penalties will only be levied on one individual, or the entire Team. Separate penalties may apply for disruption during the finals and/or for teams with no more competition runs.

The Competition Lead(s) may, at any time, question a participant on their familiarity with any relevant matter of safety, particularly if a dangerous or potentially dangerous situation is brought to their attention. They may make **immediate** penalty judgment(s), **without warning**, if they see unsafe practices, or severe/repeated violations of the code of conduct.

Strike One

- A verbal, or written, warning will be issued.

Strike Two

- Removal from current Session, or removal from the RMRC area for 4 competition hours, whichever is longer.
- Scores from the current Session will be struck from the board.
- Any **unsupervised** equipment, or Robots, physically present (e.g. on a table, in a lane, in a thoroughfare) will be impounded until the end of the next run.
- Any **unsafe** equipment, or Robots, physically present (e.g. on a table, in a lane, in a thoroughfare) will be impounded and removed from the field of competition until it has been deemed safe to return to the competition.
- RoboCup reserves the right to revoke an award if fraudulent behavior can be proven after the award ceremony takes place.

Strike Three

- Disqualification from the **entire** competition.
- Requested to leave the venue.
- If it is clear that a Team, or Team member, intentionally violates the code of conduct, the person or entire Team may be banned from future participation in RoboCup competitions.

Please post any questions and comments to the mailing list, http://list.oarkit.org/ , for group discussion.

Appendix A: Score Sheets

While all tests only involve points counting and, thus, don't require scoresheets, a specialized scoresheet can be convenient to more easily track the mission and provide a standardized record. Examples of scoresheets are available from https://rrl-rmrc.org , for both the preliminary rounds and the finals (last page). These may be folded or cut along the marked lines for easier handling during the competition.

Note that the annotations on the scoresheets are only intended as a memory aid. The full rules document is the reference for how to score a test.

It is recommended that each time a mission is reset (and the scores reset to zero) a new scoresheet is used. Where multiple scoresheets exist for

- Lap based scoresheet:
    - Incline and Center - Horizontal
    - Incline and Center - Inclined
    - Sand and Gravel
    - Ramps on Crossover Slope - Continuous Ramps
    - Ramps on Crossover Slope - Pinwheel Ramps
    - Elevated Ramps
    - Stairs
    - Align
    - K-Rails - Horizontal
    - K-Rails - Crossover Slope
    - Hurdles - Single Layer
    - Hurdles - Double Layer
- Keypad Omni scoresheet:
    - Keypad Omni
- Linear Rail Pick scoresheet:
    - Linear Rail Pick
- Linear Rail Inspect scoresheet:
    - Linear Rail Inspect
- Sensor Cabinet scoresheet:
    - Sensor Cabinet
- Labyrinth scoresheet
  Should be used with a Sensor Cabinet scoresheet. A cut down version is placed next to each Labyrinth scoresheet for convenience.
    - Labyrinth - Flat
    - Labyrinth - K-Rails

Appendix B: Changelog

2025-07-09

- Migrated scoresheets to their own file.
- Clarified autonomy bonus for the Labyrinth - it still applies for the sensor cabinet, and both QR code mapping and sensor cabinet autonomous points count towards Best-in-Class Autonomy.
- Clarified that there is only one sensor cabinet in the finals (in the dexterity field - the one in the labyrinth is removed).

2025-07-07

- Added scoresheets.
- Fixed error in note of keypad omni (which still referenced pushing a key on another keypad rather than driving out and returning).
- Replaced mentions of Sensor Cabinet with Sensor Cabinet.

2025-06-20

- Initial published version for 2025

2026-04-12

- “2 to 4 runs” in the finals for each team -> “1 to 4 runs”
- Robots need to have a distance of 30cm to the object before executing autonomous behaviour in order to be scored.
- Changed “20 QR sheets” -> “QR sheets” in labyrinth

Appendix C: Backlog for Rules Discussion

- Revise Stowed Posture rules: Should it be allowed, that teams declare their stowed posture before the competition? The stowed posture is necessary to score autonomous points, see section “Stowed Posture”
- Revise Penalty section: It should describe cooperative handling and not punishment.